

Project Name: Elevator Odyssey

Introduction

In this busy society, many people hardly have a moment to rest or relax. Even while taking the elevator, many are already thinking about their upcoming meetings, how to prepare work reports, or the looming deadlines. To help everyone find brief moments of relaxation, we propose this plan: allowing users to experience a variety of scenic views during their elevator rides, offering their minds and bodies a small but meaningful pause to catch their breath.



Features

Immersive Scenic Experiences

MR technology transforms the elevator interior into virtual environments like beaches, forests, or starry skies, offering a calming escape from stress.

Boosting Creativity

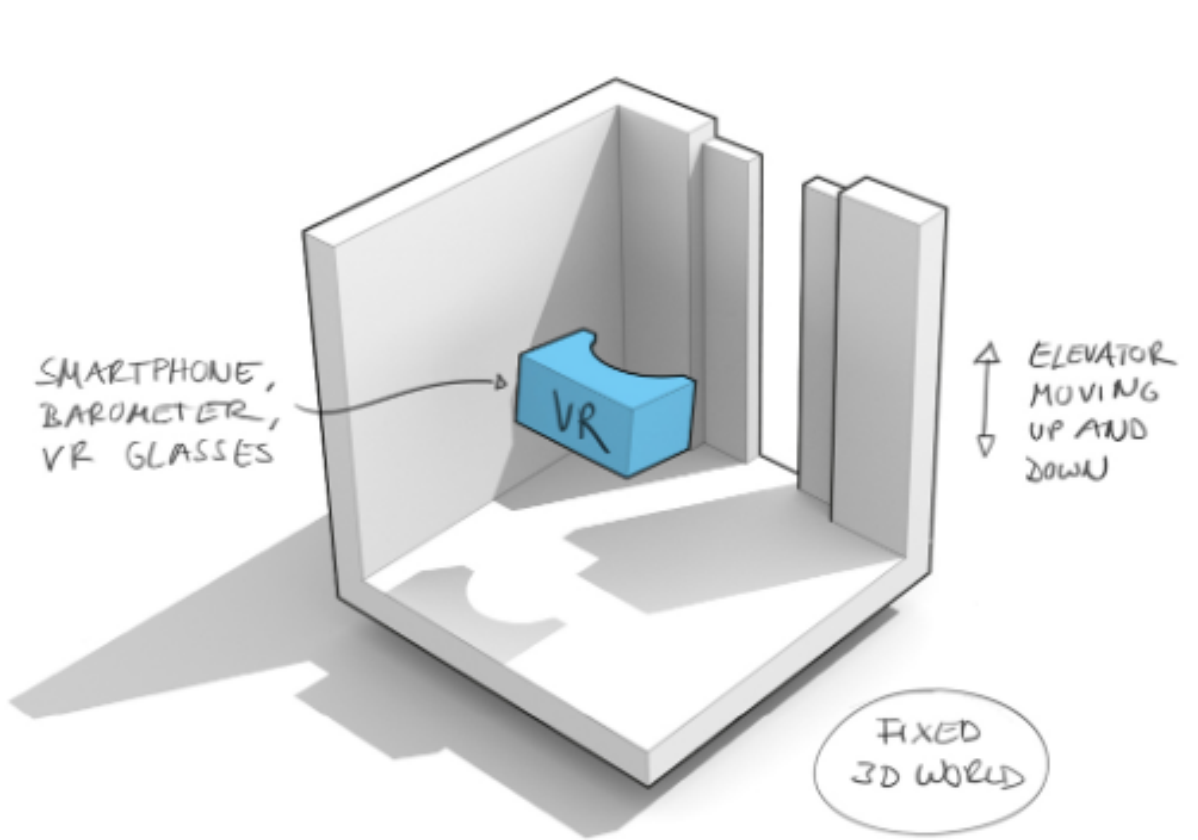
Explore art pieces or creative scenes through MR to spark fresh ideas—perfect for moments when you need inspiration.

Reducing Crowding Anxiety

MR creates the illusion of open spaces through visual effects, alleviating feelings of confinement and enhancing overall comfort, even in crowded elevators.

Usage Method

The focus of our project is on creating an immersive experience. Users only need to wear an MR headset, step into the elevator, and select their desired floor. From there, they can begin exploring the exotic cultures or breathtaking landscapes we've designed specifically for them.



Here are some experiences you can enjoy while riding the elevator:

- Relax as you select your floor. Then Immerse yourself in the unique charm of foreign cultures.
- Forget your busy schedule and give your mind and body a refreshing break.

Making Wait Time Enjoyable

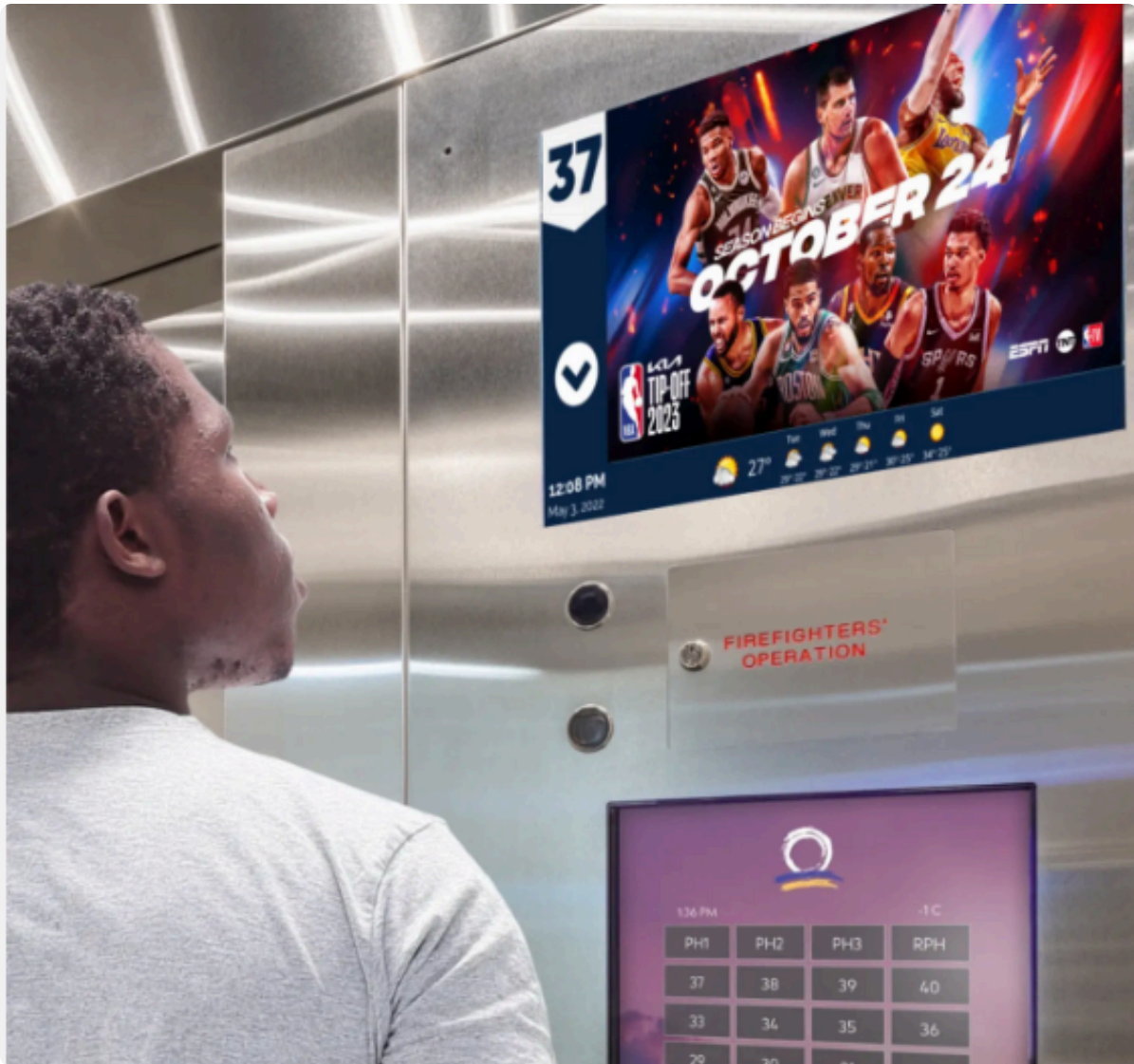
While waiting for the elevator to reach your floor, MR provides interactive content, such as mini-games or quick learning opportunities, turning dull moments into fun ones.

Personalized Experiences

MR allows users to customize their elevator ride, whether it's a tranquil beach scene or favorite videos tailored to their preferences.

A Tool for Relaxation and Meditation

MR integrates guided breathing exercises and calming visuals, helping users relax quickly and ease daily stress.



Themed floor experience

- Basement
- Desert
- Manor
- Oasis
- Grassland
- Cinema
- Space

Detect Elevator Doors Open / Close

Using Meta's built-in depth detection feature, determine the distance of objects in front of the headset. If the distance exceeds a certain value, it indicates that the elevator doors are open. Conversely, if the headset detects the elevator doors, it indicates that the doors are closed.

Virtual World Fade In / Fade Out

- Passthrough as the Background:

The real-world passthrough feed is rendered as the background of your scene using the Passthrough Layer (e.g., OVRPassthroughLayer for Meta Quest devices). This ensures that the passthrough view of the physical environment remains visible.
- Virtual Scene Objects:

Virtual objects (like 3D models or environments) are rendered on top of the passthrough background. These objects can be fully visible, partially transparent, or even invisible, depending on the desired blending effect.
- Transparency Control:

The visibility of the virtual objects is managed by adjusting the alpha (transparency) of their materials. By gradually increasing or decreasing the alpha value (e.g., Material.color.a), you can smoothly fade virtual objects in or out of view.

This creates a dynamic transition between the real-world passthrough and the virtual scene, allowing for seamless blending of the two.
- Fade In / Fade Out:

When the elevator doors open, the virtual scene gradually fades out, revealing the passthrough view of the real world. After selecting a floor button, the virtual scene corresponding to the selected floor's theme gradually fades in. Once the transparency reaches 1, the virtual scene fully covers the passthrough view and replaces it with the themed skybox for that floor.

Virtual Button Setup

Use PokeInteractable object action, to trigger the elevator moving event.

```
private PokeInteractable _elevatorbutton;
_elevatorbutton.WhenStateChanged += WhenButtonStateChanged;
private void WhenButtonStateChanged(InteractableStateChangeArgs obj)
{
    if (obj.NewState == InteractableState.Select)
    {
        // Change floor
        Elevator_switch_passthrough.instance.switch_to_virtualworld(button_id); // switch to real world
    }
}
```

Using Sky box

MR can't use skybox, because skybox will cover the passthrough. So we need to change OVRPassthroughLayer to set the skybox

```
OVRPassthroughLayer _passthroughLayer;
_passthroughLayer.projectionSurfaceType = OVRPassthroughLayer.ProjectionSurfaceType.Reconstructed;
_passthroughLayer.overlayType = OVROverlay.OverlayType.Underlay;
Camera.main.clearFlags = CameraClearFlags.SolidColor;
```

Team Member Roles and Responsibilities

In terms of task division, we've broken down the details meticulously, with a significant amount of collaboration among team members. It's therefore challenging to list tasks in a purely individual manner. The following is a general outline for reference.



312706010 徐新庭
UI: skybox, environment setup
Code: Program execution logic
Code: Other interactive elements

313554023 魏仲彦
UI: Elevator, walls, environment setup
Code: Gesture control for the elevator
Code: Other interactive elements

312553019 周彦昀
UI: skybox, environment setup
Code: Other interactive elements